

1. RSA

(a) Key generation (the receiver)

- ① Choose 2 large prime numbers of same size p and q
- ② $n = p \cdot q$
- ③ $\phi(n) = (p-1) \cdot (q-1)$
- ④ Choose $e \in \{2, 3, \dots, \phi(n)-1\}$ such that $\gcd(e, \phi(n)) = 1$
- ⑤ Compute d such that $d \cdot e = 1 \pmod{\phi(n)}$
- ⑥ Public Key: $K_p = (e, n)$ Private Key: $K_s = (d, n)$

Encryption: Use public Key: $K_p = (e, n)$

Convert the message into x where $0 \leq x < n$

$$y = E_{K_p}(x) = x^e \pmod{n}$$

Decryption: Use private Key: $K_s = (d, n)$

$$x = E_{K_s}^{-1}(y) = y^d \pmod{n}$$

(b) $p=31, q=43, e=17, x_1=9, y_2=6$

$$n = p \cdot q = 1333$$

$$\phi(n) = (p-1) \cdot (q-1) = 30 \cdot 42 = 1260$$

$$e=17, \text{ compute } d = e^{-1} \pmod{\phi(n)}$$

$$\text{E.A. } 1260 = (74) \cdot 17 + 2$$

$$17 = (8) \cdot 2 + 1 \quad \checkmark$$

$$\text{E.E.A. } 1 = 17 - 8 \cdot 2$$

$$= 17 - 8(1260 - 74 \cdot 17)$$

$$= 543 \cdot 17 - 8 \cdot 1260$$

$$d = 543$$

$$y_1 = x_1^e \pmod{n}$$

$$= 9^{17} \pmod{1333}$$

$$17 = 16 + 1$$

square: $q^1 = 9 \pmod{1333}$

$$q^2 = 81 \pmod{1333}$$

$$q^4 = (q^2)^2 = 81^2 = 6561 \pmod{1333} = 1229 \pmod{1333}$$

$$q^8 = (q^4)^2 = 1229^2 = 1510441 \pmod{1333} = 152 \pmod{1333}$$

$$q^{16} = (q^8)^2 = 152^2 = 23104 \pmod{1333} = 443 \pmod{1333}$$

Multiply:

$$\begin{aligned} 9^{17} &= 9^{16} \cdot 9^1 \\ &= 443 \cdot 9 \pmod{1333} \\ &= 3987 \pmod{1333} \\ &= 1321 \pmod{1333} \end{aligned}$$

$$y_1 = 1321$$

$$\begin{aligned} x_2 &= y_2^d \pmod{n} \\ &= 6^{593} \pmod{1333} \end{aligned}$$

$$\begin{aligned} 593 &= 0b1001010001 \\ &= 2^9 + 2^6 + 2^4 + 2^0 \\ &= 512 + 64 + 16 + 1 \end{aligned}$$

$$6^1 = 6 \pmod{1333}$$

$$6^2 = 36 \pmod{1333}$$

$$6^4 = 36^2 = 1296 \pmod{1333}$$

$$6^8 = 1296^2 = (-37)^2 = 1369 \pmod{1333} = 36 \pmod{1333}$$

$$6^{16} = 36^2 = 1296 \pmod{1333}$$

$$6^{32} = 1296^2 = 36 \pmod{1333}$$

$$6^{64} = 36^2 = 1296 \pmod{1333}$$

$$6^{128} = 1296^2 = 36 \pmod{1333}$$

$$6^{256} = 1296 \pmod{1333}$$

$$6^{512} = 36 \pmod{1333}$$

$$\begin{aligned} 6^{593} &= 6^{512} \cdot 6^{64} \cdot 6^{16} \cdot 6^1 \\ &= 36 \cdot 1296 \cdot 1296 \cdot 6 \pmod{1333} \\ &= 36 \cdot ((-37) \cdot (-37)) \cdot 6 \pmod{1333} \\ &= (36 \cdot 36) \cdot 6 \pmod{1333} \\ &= 1296 \cdot 6 \pmod{1333} \\ &= 7776 \pmod{1333} \\ &= 1111 \pmod{1333} \end{aligned}$$

2	593	
2	296	1
2	148	0
2	74	0
2	37	0
2	18	1
2	9	0
2	4	1
2	2	0
2	1	0
0	0	1

(c) No, RSA is not CPA-secure because in the encryption, the attacker can compute both plaintext's ciphertext in the experiment, they can know which message is being encrypted by comparing the ciphertexts.

2. RSA signature

(a) Key generation (signer)

① two large primes p, q

② $n = p \cdot q$

③ $\phi(n) = (p-1)(q-1)$

④ choose $e \in \{2, \dots, \phi(n)-1\}$ such that $\gcd(e, \phi(n)) = 1$

⑤ compute d such that $d \cdot e = 1 \pmod{\phi(n)}$

⑥ Public Key: (e, n) Private Key: (d, n)

Signing:

$$S = D^d \pmod{n}$$

send (D, S) to verifier

Verification:

$$D' = S^e \pmod{n}$$

Ver = 1 if $D' = D$
= 0 otherwise

b) $N=91, e=5, d=29, m=28$

$$S = m^d \pmod{N} = 28^{29} \pmod{91}$$

$$29 = 16 + 8 + 4 + 1$$

$$28^1 = 28 \pmod{91}$$

$$28^2 = 784 \pmod{91} = 56 \pmod{91}$$

$$28^4 = 56^2 \pmod{91} = 3136 \pmod{91} = 42 \pmod{91}$$

$$28^8 = 42^2 = 1764 = 35 \pmod{91}$$

$$28^{16} = 35^2 = 1225 = 42 \pmod{91}$$

$$28^{29} = 28^{16} \cdot 28^8 \cdot 28^4 \cdot 28^1$$

$$= 42 \cdot 35 \cdot 42 \cdot 28$$

$$= 42^2 \cdot 35 \cdot 28$$

$$= 35 \cdot 35 \cdot 28$$

$$= 42 \cdot 28$$

$$= 1176$$

$$= 84 \pmod{91}$$

$$S = 84$$

(c) $m' = s^e \bmod N$
 $= 84^5 \bmod 91$

$5 = 4 + 1$

$84^1 = 84 \bmod 91$

$84^2 = (-7)^2 = 49 \bmod 91$

$84^4 = 49^2 = 2401 = 35 \bmod 91$

$84^5 = 84^4 \cdot 84^1$

$= 35 \cdot 84 = 2940 = 28 \bmod 91$

$\therefore m' = 28 = m$

$\therefore \text{Vrfy} = 1$

(d) ① The key generation is identical as without a hash function

② Signing: compute a hash of the document

$h = H(x)$

$s = h^d \bmod n$

(x, s) send to the verifier

③ Verification:

compute: $h = H(x)$

$h' = s^e \bmod n$

if $h' = h$, $\text{Vrfy} = 1$

otherwise, $\text{Vrfy} = 0$

(e) ① RSA can only sign numbers smaller than n , but a document can be larger than n .

② Signing a smaller and fixed size number makes computation faster

③ Hash function has the property of collision resistant. Changing the document after it is signed will fail verification.

3. DHKE

- a) ① In the public channel, Alice and Bob agree on two values (p, g) , where p is a large prime number, and g is a generator modulo p .
- ② Alice choose a private random number a
 Bob choose a private random number b
- ③ Alice computes $A = g^a \mod p$ and send A to Bob
 Bob computes $B = g^b \mod p$ and send B to Alice
- ④ Alice: $K = B^a \mod p$
 Bob: $K = A^b \mod p$, where K is their secret key.

b) $q = 467, g = 2, x = 400, y = 134$

$$K = g^{xy} \mod q$$

$$= 2^{400 \cdot 134} \mod 467$$

$$= 2^{53600} \mod 467$$

$$53600 = (115) \cdot 466 + 10$$

$$2^{53600} = 2^{115 \cdot 466 + 10}$$

$$= 2^{115 \cdot 466} \cdot 2^{10}$$

$$\therefore 2^{466} = 1 \mod 467 \text{ (Fermat's Little Theorem)}$$

$$2^{115 \cdot 466} \cdot 2^{10} = 2^{10} \mod 467$$

$$= 1024 \mod 467$$

$$= 90$$

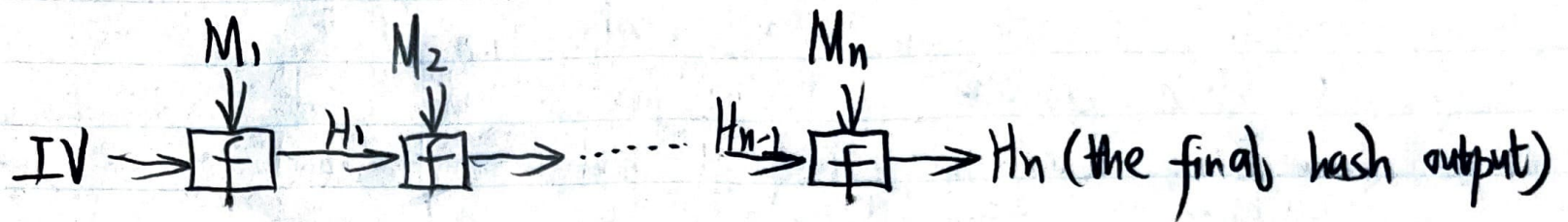
4. (a) Pigeonhole principle:

If N objects are placed into k boxes, then at least one box contains at least: $\lceil \frac{N}{k} \rceil$ objects.

(b) Birthday paradox/attack

In a group of only 23 people, the probability that at least two people have the same birthday is about 50%.

(c) Merkle-Damgård construction
a way to build a hash function that can hash any length messages using a compression function that only handle fixed-size blocks.



(d) Avalanche effect

In a good hash function, a small change in the input should cause a large, unpredictable change in the output.

(e) Properties of a good hash function

- ① Deterministic ② Fixed-length output ③ fast to compute
- ④ Preimage resistance ⑤ Second preimage resistance
- ⑥ Collision resistance ⑦ Avalanche effect

(f) popular hash functions:

MDS	SHA-1	SHA-256	SHA-384	SHA-512
output: 128 bits	160 bits	256 bits	384 bits	512 bits